

Tag & Track



Conner Museum of Vertebrates

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Project Overview

- **Objective:** Our objective is to make an iPad app that allows the employees at the Conner Museum to more easily perform their loan tracking.
- **Problem Statement:** The current way the Museum does things is all on paper, which can get hard to keep track of as well as tedious to work with. Our app aims to make this part of their job less tedious.
- **Scope:** The scope for this project includes loading specimen from arctos, allowing the creating and saving of loans, a way to track what shelf specimen are on, a way to export loan data, as well as a centralized QR code system to make everything quickly scannable.



Technologies Used





Functional Requirements

User Interface: UI must be intuitive and clear, support camera access, and remain consistent throughout.

Database: The database must be local and allow the adding a viewing of data. Further, it must be fully accessible/editable by the system.

General Functionality: The system should interface with all internal systems, as well as Arctos fluidly. Manual fallbacks should also be implemented.



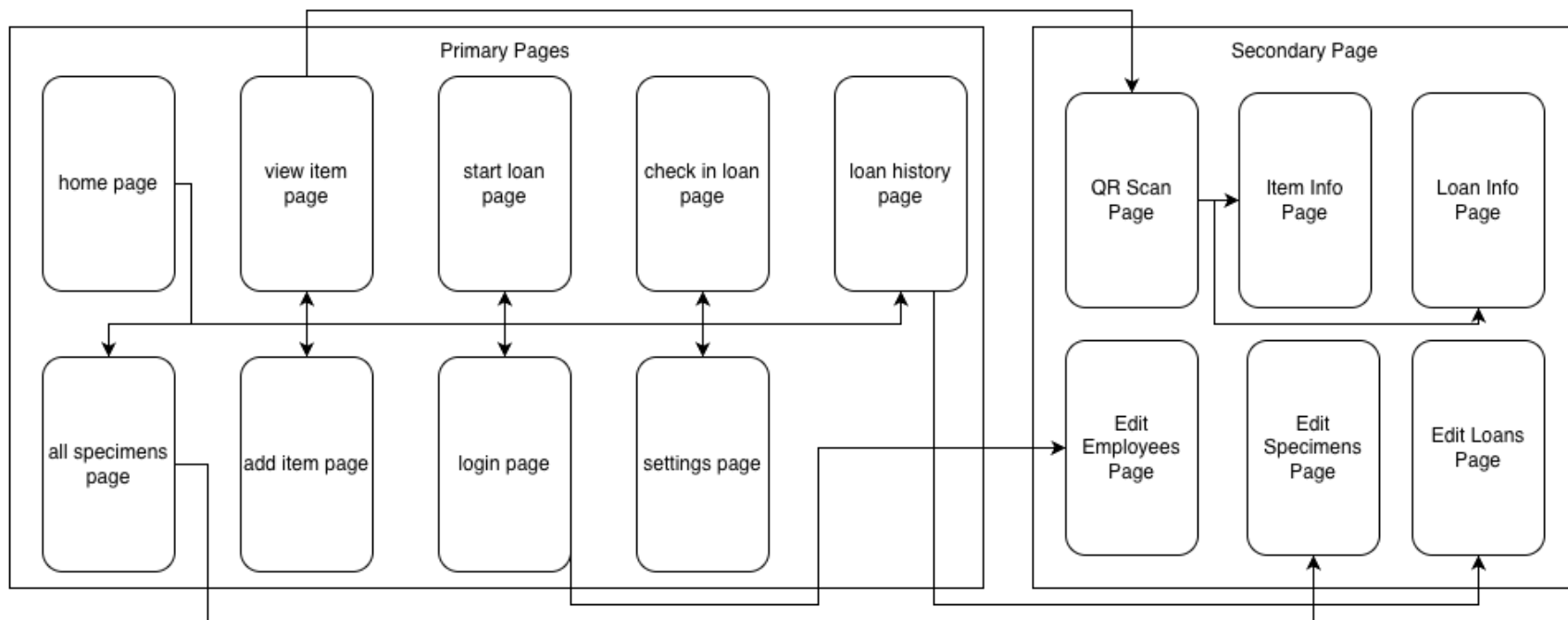
Nonfunctional Requirements

- The system shall support reduced functionality without internet access.
- The system shall support at least 5,000 specimens in the local database.
- A QR code should generate in less than 1 second
- The system shall store at least 1,000 unsynced transactions locally
- The system shall support at least 500 transactions per day without data loss



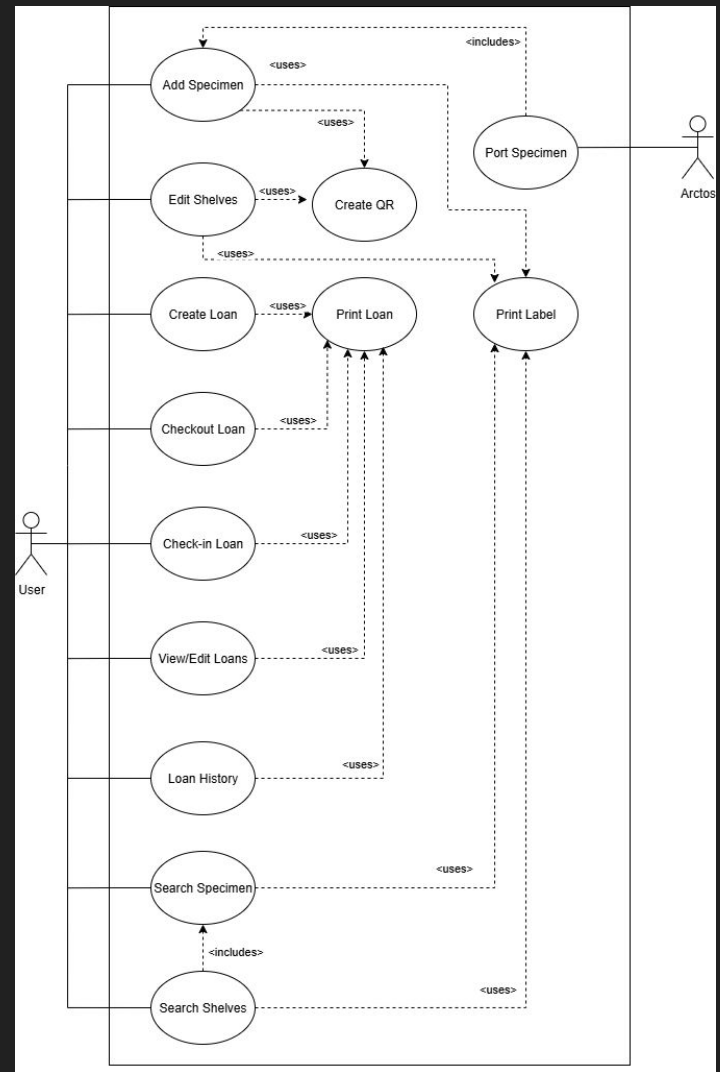
System Architecture

Tag and Track





Use Cases





Sprint 3 Achievements - Database

- Database Tables for Specimen and Loans created
- Functions for both tables created
- Initialization of the database and storing it on the device
- Beginnings of refactoring backend and UI to handle the internal database instead of the temporary one



Sprint 3 Achievements - Frontend

- UI pages more developed.
 - Headers.
 - Text entry.
 - More in depth tables.
 - Reformatting for a professional look
- UI reacts to changes made on the backend

TagAndTrack

Loan 1

Type	Loan
ID	1
Arctos ID	None
Borrower	Marcus Lee
Email	marcus.lee@wsu.edu
Checked out	2025-11-02
Due	2025-12-02
Specimens	26



Specimens in loan

ID	Name	View
90	Eagle Feather	Open
50	Beaver Skull	Open
68	Fox Pelt	Open
23	Salmon Specimen	Open
91	Lion Claw	Open



Sprint 3 Achievements - Backend

- Check in loan pipeline
- Check out loan pipeline
- Email utility

Tag and Track Loan 9402211781446263575 Confirmed ➔



tagandtrackwsu@gmail.com

to me ▼

Loan Name: My Loan

Loan ID: 9402211781446263575

Loan Description: Test

Client Name: Martin

Date Checked Out: 12/4/2025 1:22:04 PM

Date Due: 12/31/9999 11:59:59 PM

Items included in loan:

2,ARC-000002,Bear Femur,Egg sample in container,False

0,ARC-000000,Salmon Specimen,Cleaned bone specimen,False

3:08 PM Thu Dec 4



Finalize Loan

My Loan

Loan Name

Test loan

Loan Description

Martin

Client Name

Martin.hundrup@WSU.edu

Client Email



Confirm Loan

ID	Arctos ID	Name	Description	Remove Item
0	ARC-000000	Salmon Specimen	Cleaned bone specimen	Remove
1	ARC-000001	Lion Claw	Preserved specimen stored in fluid jar	Remove



Sprint 3 Retrospective

What went well

- Loan workflow integration.
- Backend/UI communication.
- UI refinement.
- Database testing.
- Client presentation and feedback gathering.

Challenges Faced

- Obtaining an API key.
- Backend changes that required frontend changes and vice versa.
- Avoiding merge conflicts.
- Development Environment issues.



System Demonstration

Demo Time!



Testing

- Testing Methodologies: So far, all testing has been done through manual and unit testing. Because of .NET MAUI, we can test both the UI and backend with an iPad, or just our Windows devices.
- Test Results: A lot of the UI has been enhanced due to our manual testing methods, as well as our backend due to our Unit testing.
- Challenges: The only challenges we have had with testing is for our database. Because the app is for an iPad, we can only test the usage of the database on an iPad device to ensure all operations are correct.



Responsible Use of AI

Usage

- AI was used at times during development of this project.

Applications and Improvements

- Referenced at times to assist in coming up with approaches to code problems, improving efficiency of certain code by at least 20%.
- Assisted in understanding UI elements and creation in .NET MAUI, shortening the research process for UI elements by 50%.

Limitations

- We would verify any code made by the AI and do thorough testing to ensure it was sound.

We understand that final decisions and deliverables are our responsibility.



Plan and Deliverables for Next Semester

Functional Requirements

- The system shall allow users to add entries to the local database.
- The system shall interface with the existing Arctos database to display specimen information.
- The system shall provide a fallback manual ID entry mode when scanning fails, with the same validation and confirmation steps.

Nonfunctional requirements

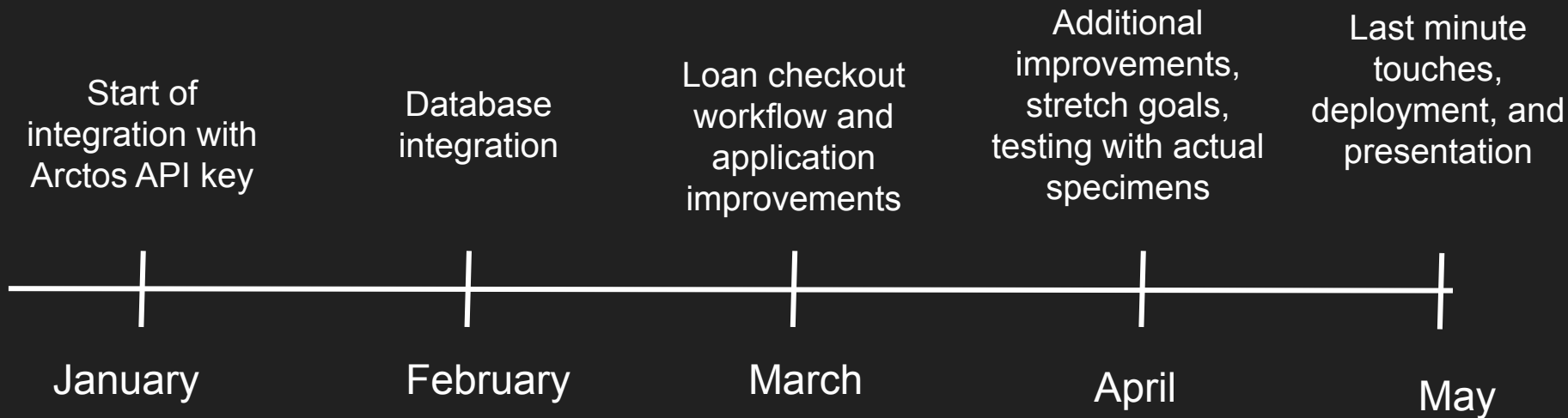
- Specimen information should load in under 2 seconds.
- All stored records shall be encrypted using AES-256, and all client-server communication shall use TLS 1.3.
- Save states should be created after every action taken.

Final Deliverables

- Loan check-in and check-out.
- Csv exporting capabilities.
- QR Code functionality with program.
- Specimen viewing without Arctos.
- Capability to list specimens located on a shelf.



Timeline for Next Semester





Conclusion

Progress

- Loan workflow has a rough implementation.
- User interface has a good number of pages thought out and will continue to be worked on.
- Database integration will be more straightforward thanks to testing performed.

Lessons

- iPad OS development required a different approach.
- Project creep can be difficult to deal with, especially with busy schedules.
- API keys can take time to obtain.

Achievements

- Better understanding and implementation of loan workflow.
- User Interface is developing and gradually looking more professional.

Thank you!